critical orbit, the red curve (the inward gravitational pull on the *Endurance*) and the blue curve (the outward centrifugal force) cross, so the two forces are in balance.

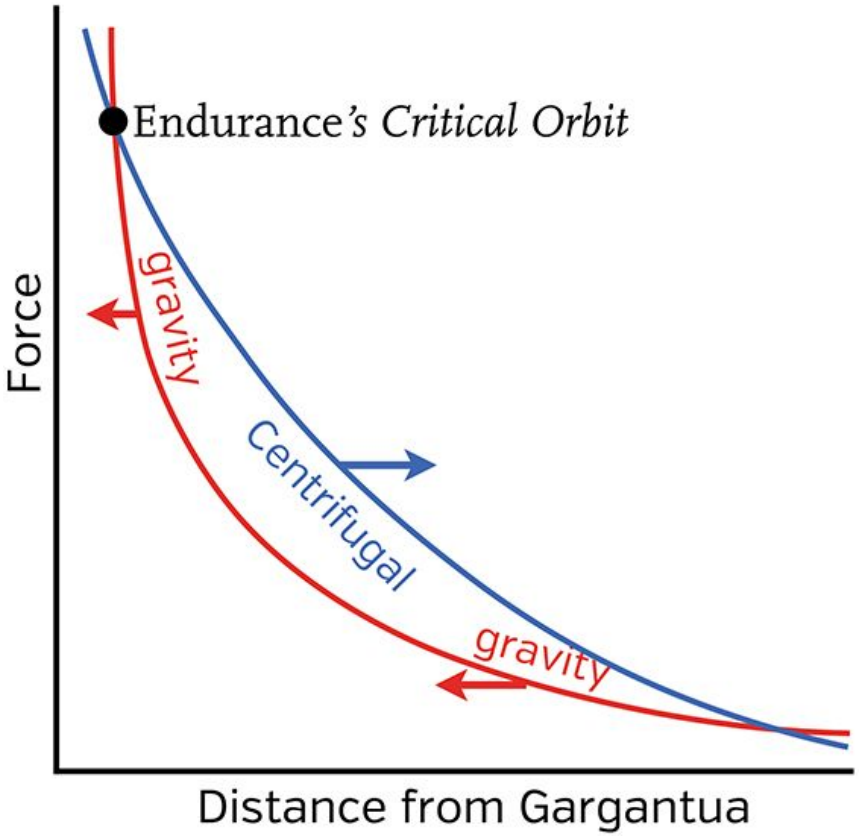


Fig. 27.6. The gravitational and centrifugal forces acting on the *Endurance*, and how they change with changing distance from Gargantua.

However, the balance is unstable, as our volcano-rim analogy suggests.[50] If the *Endurance* is randomly pushed inward just a bit, then gravity overwhelms the centrifugal force (the red curve rises above the blue curve), so the *Endurance* is pulled on inward toward Gargantua's horizon. If the *Endurance* is pushed outward just a bit, then the centrifugal force wins the battle with gravity (the blue curve is above the red curve), so the *Endurance* is pushed on outward, escaping Gargantua's tight grip.